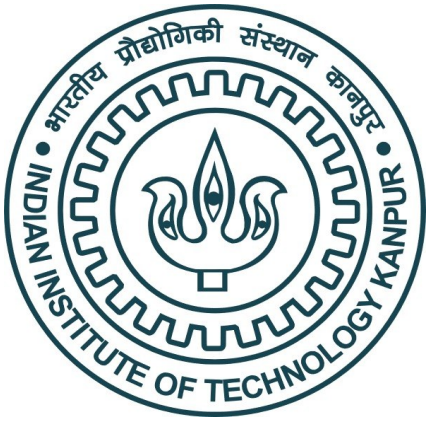
Topics in Arithmetic: Analytic Number Theory (MTH497)

Author:
Saptarshi Mandal
Department of Mathematics
IIT Kanpur

Supervisor:
Prof. Somnath Jha
Department of Mathematics
IIT Kanpur



Department of Mathematics and Statistics
Indian Institute of Technology, Kanpur

Abstract

This report follows Chapters VI and VII of Serre's *A Course in Arithmetic*. The first part proves Dirichlet's theorem on primes in arithmetic progressions.

We develop the duality theory of finite abelian groups and the orthogonality relations for Dirichlet characters. We then prove the analytic input: convergence of Dirichlet series on half-planes, Landau's theorem for non-negative coefficients, and the continuation of $\zeta(s)$ to $\Re(s) > 0$ with a simple pole at $s = 1$. Non-vanishing of $L(1, \chi)$ for $\chi \neq 1$ is deduced from the product $\zeta_m = \prod_\chi L(s, \chi)$ together with Landau's theorem. This shows that the primes $p \equiv a \pmod{m}$ have analytic density $1/\varphi(m)$, and gives the density $\frac{1}{2}$ of primes for which a non-square a is a quadratic residue.

The second part treats modular forms for $\mathrm{SL}_2(\mathbb{Z})$: the fundamental domain, Eisenstein series, and the valence formula. From these we derive the structure $\bigoplus_k M_k = \mathbb{C}[G_2, G_3]$, the dimension formula, the cusp form Δ, and the fact that every weight-0 modular function is a rational function of the modular invariant j.

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Notations

- ω — the set of integers ≥ 0.
- $P \subseteq \omega$ — the set of all prime numbers.
- $H \preceq G$ — H is a subgroup of G.
- $\langle g \rangle$ — the cyclic subgroup generated by $g \in G$.
- $U_m = G(m) = (\mathbb{Z}/m\mathbb{Z})^\times$ — the multiplicative group of invertible residue classes modulo m, of order $\varphi(m)$.
- $\mathcal{O}(U)$ — the set of functions holomorphic for $z \in U \subseteq \mathbb{C}$.
- $f_n \Rightarrow f$ — f_n converges uniformly to f.
- $\Re(z), \Im(z)$ — the real and imaginary parts of $z \in \mathbb{C}$.
- $\mathbb{H} = \{z \in \mathbb{C} : \Im(z) > 0\}$ — the Poincaré upper half-plane.
- $G = \mathrm{PSL}_2(\mathbb{Z}) = \mathrm{SL}_2(\mathbb{Z})/\{\pm 1\}$ — the modular group.
- M_k, M_k^0 — the $\mathbb{C}$-vector spaces of modular forms and cusp forms of weight $2k$.

1 Dual Groups

This part of the report follows Chapter VI of Serre's *A Course in Arithmetic* and proves **Dirichlet's theorem**: if $\gcd(a, m) = 1$, there are infinitely many primes $p \equiv a \pmod{m}$. In this section G is a finite abelian group, written multiplicatively.

Definition 1.1 (Dual group)**.** $\hat{G} := \mathrm{Hom}(G, \mathbb{C}^\times)$, is the group of *characters* of G.

If $n = |G|$ then $\chi(g)^n = 1$, so $|\chi(g)| = 1$ and $\chi(g^{-1}) = \overline{\chi(g)}$. If $G = \langle s \rangle$ is cyclic of order n, then $\chi \mapsto \chi(s)$ is an isomorphism $\hat{G} \cong \mu_n$; in particular $|\hat{G}| = |G|$.

Proposition 1.2. Let $H \preceq G$.

(a) Every character of H extends to G. Hence $1 \to \widehat{G/H} \to \hat{G} \to \hat{H} \to 1$ is exact.

(b) $|\hat{G}| = |G|$.

(c) The map $\iota : G \to \hat{\hat{G}}$, $\iota(x)(\chi) = \chi(x)$, is an isomorphism.

Proof. (a) Induct on $[G : H]$. Let $\chi \in \hat{H}$, $x \in G \smallsetminus H$, $n > 1$ minimal with $x^n \in H$, and $w \in \mathbb{C}^\times$ with $w^n = \chi(x^n)$. Define $\chi'(hx^a) = \chi(h)w^a$ on $H' = \langle H, x \rangle$. If $hx^a = h'x^{a'}$ then $x^{a'-a} \in H$, so $n \mid a' - a$ and $\chi(h'^{-1}h) = w^{a'-a}$; thus χ' is well defined, and clearly a character extending χ. Apply induction to H'. Exactness is clear: the kernel of restriction consists of the characters trivial on H.

(b) Induct on $|G|$: for a cyclic $1 \neq H \preceq G$, (a) gives $|\hat{G}| = |\hat{H}|\,|\widehat{G/H}| = |H|\,|G/H|$.

(c) By (b), $|G| = |\hat{G}| = |\hat{\hat{G}}|$, so it suffices that ι is injective. If $x \neq 1$, the character of $\langle x \rangle$ sending x to $e^{2\pi i/o(x)}$ extends to some $\chi \in \hat{G}$ with $\chi(x) \neq 1$. □

Proposition 1.3 (Orthogonality)**.** Let $n = |G|$. For $\chi \in \hat{G}$ and $x \in G$,

$$\sum_{y \in G} \chi(y) = \begin{cases} n & \chi = 1 \\ 0 & \chi \neq 1 \end{cases} \qquad \sum_{\psi \in \hat{G}} \psi(x) = \begin{cases} n & x = 1 \\ 0 & x \neq 1. \end{cases}$$

Proof. If $\chi \neq 1$, pick y_0 with $\chi(y_0) \neq 1$; then $\chi(y_0) \sum_y \chi(y) = \sum_y \chi(y_0 y) = \sum_y \chi(y)$, so the sum vanishes. The second formula is the first applied to $\hat{G}$ and the character $\iota(x)$, which is non-trivial iff $x \neq 1$ by Proposition 1.2 (c). □

1.1 Dirichlet characters

Let $U_m = (\mathbb{Z}/m\mathbb{Z})^\times$, of order $\varphi(m)$.

Definition 1.4 (Dirichlet character)**.** A *character modulo* m is an element $\chi \in \hat{U}_m$. It is viewed as a function on $\mathbb{Z}$ by $\chi(a) = \chi(a \bmod m)$ if $\gcd(a, m) = 1$ and 0 otherwise; it is then completely multiplicative and m-periodic.

Example. For $m = 4$ the non-trivial character is $\chi_4(n) = (-1)^{(n-1)/2}$ for n odd, 0 for n even. For $m = 8$, with $\varepsilon(x) \equiv \frac{x-1}{2}$ and $\omega(x) \equiv \frac{x^2-1}{8} \pmod 2$, the non-trivial characters are $(-1)^{\varepsilon}$, $(-1)^{\omega}$, $(-1)^{\varepsilon+\omega}$.

Proposition 1.5. Let $a \neq 0$ be square-free and $m = 4|a|$. There is a unique character χ_a modulo m with $\chi_a(p) = \left(\frac{a}{p}\right)$ for all primes $p \nmid m$. Moreover $\chi_a^2 = 1$, and $\chi_a \neq 1$ if $a \neq 1$.

Proof. Every class in U_m contains a product of primes $p \nmid m$, which gives uniqueness and $\chi_a^2 = 1$. For existence, let first $a = l_1 \cdots l_k > 0$ with l_i distinct odd primes. By the Chinese remainder theorem $U_m \cong U_4 \times U_{l_1} \times \cdots \times U_{l_k}$, and

$$\chi_a(x) = (-1)^{\varepsilon(x)\varepsilon(a)} \left(\frac{x}{l_1}\right) \cdots \left(\frac{x}{l_k}\right)$$

is a character with values ± 1. Quadratic reciprocity gives $\left(\frac{p}{l_i}\right) = (-1)^{\varepsilon(p)\varepsilon(l_i)} \left(\frac{l_i}{p}\right)$, and $\varepsilon(a) \equiv \sum \varepsilon(l_i)$, so $\chi_a(p) = \left(\frac{a}{p}\right)$. It is non-trivial since, by the Chinese remainder theorem, some x is a non-residue mod l_1 and $\equiv 1$ modulo $4l_2 \cdots l_k$. The remaining cases $a = -b$, $\pm 2b$ are handled in the same way, using $\left(\frac{-1}{p}\right) = (-1)^{\varepsilon(p)}$ and $\left(\frac{2}{p}\right) = (-1)^{\omega(p)}$. □

2 Some results from Complex Analysis

We recall some results from complex analysis (Serre, Chapter VI, § 2.1) needed for the analytic theory of Dirichlet series. Let U be any open subset of $\mathbb{C}$. **Notation.** $\mathcal{O}(U)$ denotes the set of all holomorphic functions on U.

Proposition 2.1 (Weierstrass)**.** Let $\langle f_n : n < \omega \rangle$ be a sequence in $\mathcal{O}(U)$ such that $f_n \Rightarrow f$ on every compact set $K \subseteq U$. Then $f \in \mathcal{O}(U)$ and $f_n' \Rightarrow f'$ on every compact set $K \subseteq U$.

Proof. Since holomorphicity and uniform convergence on compact sets are local properties, let $D = \overline{D(z_0, R)} \subseteq U$ be any closed disc contained in U, and let $C = \partial D$ be its boundary circle oriented counterclockwise. As $f_n \Rightarrow f$ uniformly on the compact set D, the limit f is continuous on D. For any point z in the interior $D(z_0, R)$, Cauchy's integral formula gives

$$f_n(z) = \frac{1}{2\pi i} \int_C \frac{f_n(\zeta)}{\zeta - z}\, d\zeta.$$

For z restricted to a smaller closed concentric disc $D_r = \overline{D(z_0, r)}$ with $r < R$, we have $|\zeta - z| \geq R - r > 0$ for all $\zeta \in C$. Thus $\frac{f_n(\zeta)}{\zeta - z} \Rightarrow \frac{f(\zeta)}{\zeta - z}$ uniformly for $\zeta \in C$. Passing to the limit under the integral sign yields

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z}\, d\zeta \quad \text{for all } z \in D(z_0, R),$$

which is a integral with continuous integrand, hence holomorphic in $D(z_0, R)$. Thus $f \in \mathcal{O}(U)$.

For the derivatives, Cauchy's differentiation formula gives, for $z \in D_r$,

$$f_n'(z) - f'(z) = \frac{1}{2\pi i} \int_C \frac{f_n(\zeta) - f(\zeta)}{(\zeta - z)^2}\, d\zeta \Rightarrow |f_n'(z) - f'(z)| \leq \frac{R}{(R-r)^2} \sup_{\zeta \in C} |f_n(\zeta) - f(\zeta)|.$$

Since $\sup_{\zeta \in C} |f_n(\zeta) - f(\zeta)| \to 0$, we have $f_n' \Rightarrow f'$ uniformly on D_r. Since any compact set $K \subseteq U$ can be covered by finitely many such closed discs D_r, $f_n' \Rightarrow f'$ uniformly on K. □

Corollary 2.2. For every $k < \omega$, $f_n^{(k)} \Rightarrow f^{(k)}$ on every compact set $K \subseteq U$.

Lemma 2.3 (Abel's Summation by Parts)**.** Let $\langle a_n : n < \omega \rangle$, $\langle b_n : n < \omega \rangle$ be two sequences in $\mathbb{C}$. Define:

$$A_{k,m} = \sum_{n=k}^{m} a_n \quad \text{and} \quad S_{k,m} = \sum_{n=k}^{m} a_n b_n.$$

Then we have:

$$S_{k,m} = \sum_{n=k}^{m-1} A_{k,n}(b_n - b_{n+1}) + A_{k,m} b_m.$$

Proof. Setting $A_{k,k-1} := 0$, we can write $a_n = A_{k,n} - A_{k,n-1}$ for every $k \leq n \leq m$. Substituting this into $S_{k,m}$ and shifting the index of the second sum gives

$$S_{k,m} = \sum_{n=k}^{m}(A_{k,n} - A_{k,n-1})b_n = \sum_{n=k}^{m} A_{k,n}b_n - \sum_{n=k}^{m-1} A_{k,n}b_{n+1} = \sum_{n=k}^{m-1} A_{k,n}(b_n - b_{n+1}) + A_{k,m}b_m. \quad \square$$

Lemma 2.4. Let $\alpha, \beta \in \mathbb{R}$ with $0 \leq \alpha < \beta$ and $z = x + iy$ with $x, y \in \mathbb{R}$ and $x > 0$. Then

$$|e^{-\alpha z} - e^{-\beta z}| \leq \left|\frac{z}{x}\right| (e^{-\alpha x} - e^{-\beta x}).$$

Proof. By the Fundamental Theorem of Calculus applied to $t \mapsto -e^{-tz}$, we have

$$e^{-\alpha z} - e^{-\beta z} = z \int_{\alpha}^{\beta} e^{-tz}\, dt.$$

Taking absolute values and using $|e^{-tz}| = e^{-\Re(tz)} = e^{-tx}$ for $t \in [\alpha, \beta]$,

$$|e^{-\alpha z} - e^{-\beta z}| \leq |z| \int_{\alpha}^{\beta} |e^{-tz}|\, dt = |z| \int_{\alpha}^{\beta} e^{-tx}\, dt = \frac{|z|}{x}(e^{-\alpha x} - e^{-\beta x}) = \left|\frac{z}{x}\right| (e^{-\alpha x} - e^{-\beta x}),$$

since $x > 0$. $\square$

3 Dirichlet Series

Let λ_n be an increasing sequence in $\mathbb{R}_{\geq 0}$ tending to ∞ (after dropping finitely many terms).

Definition 3.1 (Dirichlet series)**.** A *Dirichlet series* is $f(z) = \sum_{n \geq 0} a_n e^{-\lambda_n z}$ with $a_n \in \mathbb{C}$.

Example. 1. For $\lambda_n = \log n$ this is an *ordinary* Dirichlet series $\sum a_n n^{-s}$, $s = \sigma + it$.

2. For $\lambda_n = n$ it is a power series in e^{-z}.

A power series converges on a disc, a Dirichlet series on a right half-plane. The key fact (Serre, VI.2.2, Proposition 6) is that convergence at one point gives uniform convergence on closed sectors.

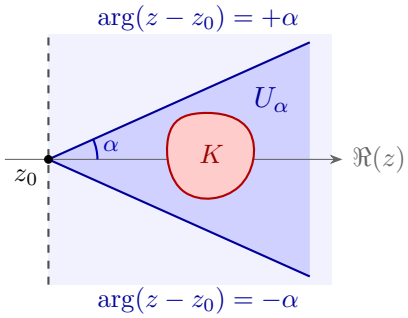


Figure 1: The closed sector (Stolz cone) U_α of half-angle $\alpha < \pi/2$ with vertex z_0. Every compact subset $K \subset \{z \in \mathbb{C} : \Re(z) > \Re(z_0)\}$ is contained in U_α for $\alpha < \pi/2$ sufficiently close to $\pi/2$.

Proposition 3.2. If f converges at z_0, it converges uniformly on $U_\alpha = \{z : \Re(z - z_0) \geq 0,\ |\arg(z - z_0)| \leq \alpha\}$ for every $\alpha < \frac{\pi}{2}$.

Proof. Take $z_0 = 0$ (replace a_n by $a_n e^{-\lambda_n z_0}$). Given $\varepsilon > 0$ choose N with $|A_{k,m}| \leq \varepsilon$ for $k, m \geq N$. For $z = x + iy \in U_\alpha \setminus \{0\}$, Abel's lemma, Lemma 2.4 and telescoping give

$$\left|\sum_{n=k}^{m} a_n e^{-\lambda_n z}\right| \leq \varepsilon e^{-\lambda_m x} + \varepsilon \left|\frac{z}{x}\right| \left(e^{-\lambda_k x} - e^{-\lambda_m x}\right) \leq \varepsilon(1 + \sec\alpha). \qquad \square$$

Corollary 3.3. If f converges at z_0, then it converges and is holomorphic on $\Re(z) > \Re(z_0)$, and may be differentiated termwise: $f^{(p)}(z) = \sum a_n(-\lambda_n)^p e^{-\lambda_n z}$.

Proof. A compact subset of $\{\Re z > \Re z_0\}$ lies in some U_α with $\alpha < \frac{\pi}{2}$. Apply Proposition 3.2 and Proposition 2.1. $\square$

Hence the convergence region contains a maximal open half-plane $\Re(z) > \rho$, and ρ is the *abscissa of convergence*. Similarly ρ^+ is the abscissa of *absolute* convergence; $\rho \leq \rho^+$. For ordinary Dirichlet series $\rho^+ \leq \rho + 1$; for instance $\eta(s) = \sum (-1)^{n-1} n^{-s}$ and $L(s, \chi_4)$ have $\rho = 0$, $\rho^+ = 1$ (see Figure 2 and Proposition 3.6).

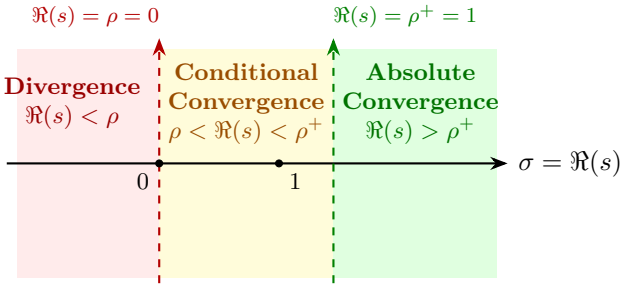


Figure 2: The three regimes of convergence in the complex s-plane for a Dirichlet series.

For non-negative coefficients, the abscissa of convergence is always a singularity of f on the real axis (Serre, VI.2.3, Proposition 7).

Theorem 3.4 (Landau)**.** Let $a_n \geq 0$ and suppose f converges for $\Re(z) > \rho \in \mathbb{R}$. If f extends holomorphically to a neighbourhood of ρ, then f converges on $\Re(z) > \rho - \varepsilon$ for some $\varepsilon > 0$.

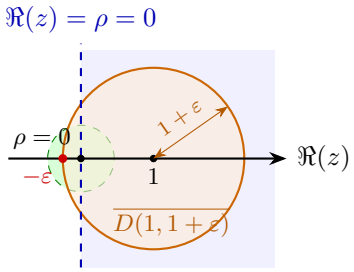


Figure 3: Geometric setup in the proof of Landau's Theorem (Theorem 3.4)

Proof. Take $\rho = 0$. Then f is holomorphic on a neighbourhood of $\overline{D(1, 1+\varepsilon)}$ for some $\varepsilon > 0$, so its Taylor series at 1 converges at $-\varepsilon$. By Corollary 3.3, $f^{(p)}(1) = (-1)^p \sum_n \lambda_n^p a_n e^{-\lambda_n}$, hence

$$f(-\varepsilon) = \sum_{p \geq 0} \frac{(1+\varepsilon)^p}{p!} \sum_n \lambda_n^p a_n e^{-\lambda_n}.$$

All terms are ≥ 0, so we may interchange the sums: $f(-\varepsilon) = \sum_n a_n e^{-\lambda_n} e^{(1+\varepsilon)\lambda_n} = \sum_n a_n e^{\lambda_n \varepsilon}$. Thus the series converges at $-\varepsilon$, hence on $\Re(z) > -\varepsilon$. □

Proposition 3.5. If (a_n) is bounded, then $\sum a_n n^{-z}$ converges absolutely for $\Re(z) > 1$.

Proposition 3.6. If the partial sums $A_{k,m} = \sum_{n=k}^m a_n$ are uniformly bounded, then $\sum a_n n^{-z}$ converges for $\Re(z) > 0$.

Proof. The first is comparison with $\sum n^{-x}$. For the second, by Corollary 3.3 it suffices to take $z = x > 0$ real. If $|A_{k,m}| \leq M$, Abel's lemma and the monotonicity of n^{-x} give $\left|\sum_{n=k}^m a_n n^{-x}\right| \leq M\left(m^{-x} + \sum_{n=k}^{m-1}(n^{-x} - (n+1)^{-x})\right) = Mk^{-x} \to 0$. □

4 The Riemann-Zeta Function

A function $f : \mathbb{N}_{\geq 1} \to \mathbb{C}$ is *multiplicative* if $f(1) = 1$ and $f(mn) = f(m)f(n)$ whenever $\gcd(m, n) = 1$. If $f(mn) = f(m)f(n)$ for *all* $m, n \geq 1$, then f is *strictly multiplicative*.

Example. Important multiplicative functions in number theory include:

(i) *Strictly multiplicative*: power function $f(n) = n^k$, and any Dirichlet character $\chi(n)$ mod m.

(ii) *Multiplicative (but not strictly)*: Euler's totient function $\varphi(n)$ (e.g., $\varphi(4) \neq \varphi(2)\varphi(2)$).

In this section, we shall assume that f is a bounded multiplicative function. By $P \subseteq \omega$, we denote the set of all prime numbers. The following fundamental result of Euler (Serre, VI.3.1, Lemmas 4 and 5) translates the unique prime factorization of integers into an infinite product factorization of the Dirichlet series of f.

Lemma 4.1 (Euler Product)**.** If f is a bounded multiplicative function, then $\sum_{n=1}^{\infty} f(n)n^{-s}$ converges absolutely for $\Re(s) > 1$, and in this domain

$$\sum_{n=1}^{\infty} f(n)n^{-s} = \prod_{p\in P}\left(\sum_{m=0}^{\infty} f(p^m)p^{-ms}\right).$$

Proof. Since f is bounded, by Proposition 3.5, the series $\sum_{n=1}^{\infty} f(n)n^{-s}$ converges absolutely for $\Re(s) > 1$. Let $S \subseteq P$ be any finite set of primes, and define

$$N(S) := \{n \geq 1 : (\forall p \in P)(p \mid n \Rightarrow p \in S)\},$$

the set of all positive integers ≥ 1 whose prime factors all belong to S. Since every $n \in N(S)$ can be written uniquely as a finite product $n = \prod_{p\in S} p^{m_p}$ with $m_p \geq 0$. Since f is multiplicative and the prime powers p^{m_p} are pairwise coprime, $f(n)n^{-s} = \prod_{p\in S} f(p^{m_p})p^{-m_p s}$. Because each series $\sum_{m=0}^{\infty} f(p^m)p^{-ms}$ converges absolutely for $\Re(s) > 1$, multiplying out the finite product of absolutely convergent series over $p \in S$ yields

$$\prod_{p\in S}\left(\sum_{m=0}^{\infty} f(p^m)p^{-ms}\right) = \sum_{n\in N(S)} f(n)n^{-s}.$$

Now, if S contains all primes $\leq X$, then every integer $1 \leq n \leq X$ belongs to $N(S)$, so

$$\left|\sum_{n=1}^{\infty} f(n)n^{-s} - \prod_{p\in S}\left(\sum_{m=0}^{\infty} f(p^m)p^{-ms}\right)\right| = \left|\sum_{n\notin N(S)} f(n)n^{-s}\right| \leq \sum_{n>X} |f(n)|\, n^{-\Re(s)}.$$

Since $\sum_{n=1}^{\infty} |f(n)|n^{-\Re(s)} < \infty$ for $\Re(s) > 1$, the tail sum on the right tends to 0 as $X \to \infty$ (i.e., as $S \uparrow P$). Thus the infinite product converges and equals $\sum_{n=1}^{\infty} f(n)n^{-s}$. □

Lemma 4.2. If f is a bounded strictly multiplicative function, then for $\Re(s) > 1$,

$$\sum_{n=1}^{\infty} f(n)n^{-s} = \prod_{p\in P} \frac{1}{1 - f(p)p^{-s}}.$$

Proof. Since f is strictly multiplicative, $f(p^m) = f(p)^m$ for all $m \geq 0$ (and $|f(p)| \leq 1$ because f is bounded). Thus for each prime p and $\Re(s) > 1$, we have $|f(p)p^{-s}| \leq p^{-\Re(s)} \leq 2^{-\Re(s)} < 1$, so the inner sum in Lemma 4.1 is a convergent geometric series:

$$\sum_{n\in N(S)} f(n)n^{-s} = \prod_{p\in S}\sum_{m=0}^{\infty} f(p^m)p^{-ms} = \prod_{p\in S}\sum_{m=0}^{\infty} \left(f(p)p^{-s}\right)^m = \prod_{p\in S} \frac{1}{1 - f(p)p^{-s}}.$$

Letting $S \uparrow P$ gives the required Euler product. □

Taking $f \equiv 1$ gives the **Riemann Zeta function**. By Lemma 4.2, for $\Re(s) > 1$,

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{p\in P} \frac{1}{1 - p^{-s}}. \tag{1}$$

Proposition 4.3 (Analytic Continuation to $\Re(s) > 0$)**.** The following are true about ζ:

(a) $\zeta(s)$ is holomorphic and $\neq 0$ for $\Re(s) > 1$.

(b) $\zeta(s) = \frac{1}{s-1} + \psi(s)$, where $\psi(s)$ is holomorphic for $\Re(s) > 0$.

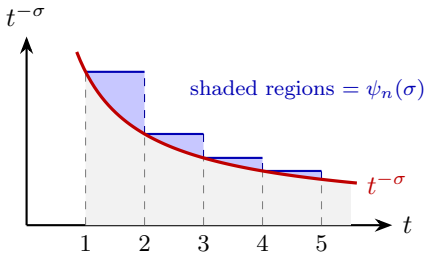


Figure 4: For real $\sigma > 1$, $\zeta(\sigma) - \frac{1}{\sigma-1} = \sum_n \psi_n(\sigma)$ is the total shaded area between the step function $n^{-\sigma}$ and the curve $t^{-\sigma}$.

Proof. (a) For $\Re(s) > 1$, $\zeta(s)$ converges absolutely by Proposition 3.5, hence is holomorphic by Corollary 3.3. To see $\zeta(s) \neq 0$, fix s with $x = \Re(s) > 1$, let S_N be the set of the first N primes and p_N the largest of them. By unique factorization and the proof of Lemma 4.2,

$$\zeta(s) \prod_{p \in S_N} (1 - p^{-s}) = \sum_{\substack{n \geq 1 \\ p \nmid n \ \forall p \in S_N}} n^{-s} = 1 + \sum_{\substack{n > p_N \\ p \nmid n \ \forall p \in S_N}} n^{-s}.$$

Taking the reverse triangle inequality, for N large enough so that $\sum_{n > p_N} n^{-x} \leq \frac{1}{2}$, we obtain

$$|\zeta(s)| \prod_{p \in S_N} (1 + p^{-x}) \geq \left| \zeta(s) \prod_{p \in S_N} (1 - p^{-s}) \right| \geq 1 - \sum_{n > p_N} n^{-x} \geq \frac{1}{2} > 0,$$

which implies $\zeta(s) \neq 0$.

(b) For $\Re(s) > 1$, note that $\frac{1}{s-1} = \int_1^\infty t^{-s}\, dt = \sum_{n \geq 1} \int_n^{n+1} t^{-s}\, dt$. Subtracting this from $\zeta(s) = \sum_{n \geq 1} n^{-s} = \sum_{n \geq 1} \int_n^{n+1} n^{-s}\, dt$ (see Figure 4), we have

$$\zeta(s) = \frac{1}{s-1} + \sum_{n=1}^{\infty} \psi_n(s), \quad \text{where} \quad \psi_n(s) = \int_n^{n+1} (n^{-s} - t^{-s})\, dt.$$

Let $\psi(s) = \sum_{n \geq 1} \psi_n(s)$. Each ψ_n is entire (an integral over a compact interval of an entire function of s), so by Proposition 2.1 it suffices to show that $\sum \psi_n$ converges uniformly on every compact $K \subseteq \{\Re(s) > 0\}$.

For any $t \in [n, n+1]$ and $s = x + iy$ with $x = \Re(s) > 0$, the derivative of $u \mapsto u^{-s}$ is $-su^{-s-1}$. By the Fundamental Theorem of Calculus,

$$|n^{-s} - t^{-s}| = \left| \int_n^t (-su^{-s-1})\, du \right| \leq (t - n) \sup_{n \leq u \leq n+1} \frac{|s|}{u^{x+1}} \leq \frac{|s|}{n^{x+1}}.$$

Integrating with respect to $t \in [n, n+1]$ gives the bound

$$|\psi_n(s)| \leq \int_n^{n+1} |n^{-s} - t^{-s}|\, dt \leq \frac{|s|}{n^{x+1}}.$$

Now let $K \subseteq \{s \in \mathbb{C} : \Re(s) > 0\}$ be any compact set. Since K is compact, there exist $\varepsilon > 0$ and $M > 0$ such that $\Re(s) = x \geq \varepsilon$ and $|s| \leq M$ for all $s \in K$. Then for all $s \in K$,

$$\sum_{n=1}^{\infty} |\psi_n(s)| \leq M \sum_{n=1}^{\infty} \frac{1}{n^{1+\varepsilon}} < \infty.$$

By the Weierstrass M-test, $\psi(s) = \sum_{n=1}^{\infty} \psi_n(s)$ converges uniformly on K, and hence defines a holomorphic function on $\{s \in \mathbb{C} : \Re(s) > 0\}$. □

Corollary 4.4. $\zeta(s)$ extends meromorphically to $\Re(s) > 0$ with a single simple pole at $s = 1$ with residue 1.

By the simple pole of $\zeta(s)$ at $s = 1$, we obtain Euler's analytic proof of the infinitude of primes and the asymptotic growth of $\sum_{p \in P} p^{-s}$ as $s \to 1^+$ (Serre, VI.3.2, Corollary 2).

Corollary 4.5. When $s \to 1^+$ ($s \in \mathbb{R}, s > 1$), one has

$$\sum_{p \in P} \frac{1}{p^s} \sim \log\left(\frac{1}{s-1}\right),$$

and the higher prime-power sum $\sum_{p \in P,\, k \geq 2} \frac{1}{kp^{ks}}$ remains bounded for $s > 1$.

Proof. For real $s > 1$, $\zeta(s) > 1$ and $0 < p^{-s} < 1$. Taking the real logarithm of the Euler product (1) and using the Taylor expansion $-\log(1-u) = \sum_{k=1}^{\infty} \frac{u^k}{k}$ for $|u| < 1$, we obtain

$$\log \zeta(s) = \sum_{p \in P} \log\left(\frac{1}{1-p^{-s}}\right) = \sum_{p \in P} \sum_{k=1}^{\infty} \frac{1}{kp^{ks}} = \sum_{p \in P} \frac{1}{p^s} + \Theta(s),$$

where $\Theta(s) = \sum_{p \in P} \sum_{k=2}^{\infty} \frac{1}{kp^{ks}}$ (rearrangement is justified since all terms are positive). For $s > 1$, we can bound $\Theta(s)$ uniformly by summing the geometric series over $k \geq 2$:

$$\begin{aligned} 0 < \Theta(s) &\leq \sum_{p \in P} \sum_{k=2}^{\infty} \frac{1}{p^{ks}} = \sum_{p \in P} \frac{1}{p^s(p^s-1)} \leq \sum_{p \in P} \frac{1}{p(p-1)} \\ &\leq \sum_{n=2}^{\infty} \frac{1}{n(n-1)} = \sum_{n=2}^{\infty} \left(\frac{1}{n-1} - \frac{1}{n}\right) = 1. \end{aligned}$$

On the other hand, by Proposition 4.3 (b), $(s-1)\zeta(s) = 1 + (s-1)\psi(s) \to 1$ as $s \to 1^+$, so

$$\log \zeta(s) = \log\left(\frac{1}{s-1}\right) + \log\big((s-1)\zeta(s)\big) = \log\left(\frac{1}{s-1}\right) + o(1) \quad \text{as } s \to 1^+.$$

Since $\Theta(s) = O(1)$ is bounded while $\log \frac{1}{s-1} \to +\infty$, dividing by $\log \frac{1}{s-1}$ yields $\sum_{p \in P} p^{-s} \sim \log \frac{1}{s-1}$. □

Remark. ζ in fact extends meromorphically to all of $\mathbb{C}$ with a single simple pole at $s = 1$, and $\xi(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$ satisfies $\xi(s) = \xi(1-s)$. The Riemann Hypothesis asserts that all non-trivial zeros lie on $\Re(s) = \frac{1}{2}$. We only need $\Re(s) > 0$.

5 L-functions

Let $m \geq 1$ be a fixed integer, and let $\chi \in \hat{U}_m$ be a Dirichlet character modulo m (extended to $\mathbb{Z}$ as in **Definition 1.4**). Clearly $\chi(n) \neq 0 \Leftrightarrow \gcd(n, m) = 1$.

Definition 5.1 (Dirichlet L-function)**.** The *Dirichlet L-function* associated to $\chi \in \hat{U}_m$ is defined by the ordinary Dirichlet series

$$L(s, \chi) = \sum_{n \geq 1} \frac{\chi(n)}{n^s}.$$

For the principal character $\chi = 1$ (where $1(n) = 1$ if $\gcd(n, m) = 1$ and 0 if $\gcd(n, m) > 1$), we get an analogue of the Riemann Zeta function with the prime factors of m removed. The next result (Serre, VI.3.3, Proposition 11) establishes that $L(s, 1)$ differs from $\zeta(s)$ only by a finite Euler factor, and therefore also has a simple pole at $s = 1$.

Proposition 5.2. For the principal character $\chi = 1$ modulo m, one has

$$L(s, 1) = \prod_{p \nmid m} \frac{1}{1 - p^{-s}} = F(s)\, \zeta(s), \quad \text{where} \quad F(s) = \prod_{p \mid m} (1 - p^{-s}).$$

In particular, $L(s, 1)$ extends to a function holomorphic for $\Re(s) > 0$, except for a simple pole at $s = 1$ with residue $\prod_{p \mid m}(1 - p^{-1}) = \frac{\varphi(m)}{m} \neq 0$.

Proof. Note that $\chi = 1$ is strictly multiplicative and bounded ($|\chi(n)| \leq 1$). By Lemma 4.2, and the fact that $p \mid m \Rightarrow \chi(p) = 0$ while $p \nmid m \Rightarrow \chi(p) = 1$, we have for $\Re(s) > 1$:

$$\begin{aligned} L(s, 1) = \sum_{n \geq 1} \chi(n) n^{-s} &= \prod_{p \in P} \frac{1}{1 - \chi(p) p^{-s}} = \prod_{p \nmid m} \frac{1}{1 - p^{-s}} \\ &= \left(\prod_{p \mid m} (1 - p^{-s}) \right) \prod_{p \in P} \frac{1}{1 - p^{-s}} = F(s)\, \zeta(s). \end{aligned}$$

Since m has only finitely many prime factors, $F(s) = \prod_{p \mid m}(1 - p^{-s})$ is a finite product of entire functions, hence is entire on $\mathbb{C}$, with $F(1) = \prod_{p \mid m}(1 - \frac{1}{p}) = \frac{\varphi(m)}{m} > 0$. By Proposition 4.3 (b), $\zeta(s)$ is holomorphic on $\{\Re(s) > 0\} \setminus \{1\}$ with a simple pole at $s = 1$, so the second assertion holds immediately. □

When $\chi \neq 1$, oscillatory cancellation among the roots of unity $\chi(n)$ improves the abscissa of convergence from $\rho = 1$ to $\rho \leq 0$ (Serre, VI.3.3, Proposition 12):

Proposition 5.3. For $\chi \neq 1$, the series $L(s, \chi)$ converges (and defines a holomorphic function) in the open half-plane $\Re(s) > 0$, and converges absolutely in the half-plane $\Re(s) > 1$. Moreover, for $\Re(s) > 1$, one has the Euler product

$$L(s, \chi) = \prod_{p \in P} \frac{1}{1 - \chi(p) p^{-s}} = \prod_{p \nmid m} \frac{1}{1 - \chi(p) p^{-s}}.$$

Proof. Since $|\chi(n)| \leq 1$ and χ is strictly multiplicative, absolute convergence and the Euler product expansion for $\Re(s) > 1$ follow directly from Proposition 3.5 and Lemma 4.2.

To show convergence of the Dirichlet series $L(s,\chi) = \sum_{n\geq 1} \chi(n)n^{-s}$ for $\Re(s) > 0$, by Proposition 3.6 it suffices to verify that the partial sums $A_{u,v} = \sum_{n=u}^{v} \chi(n)$ $(1 \leq u \leq v)$ are uniformly bounded. Since χ is periodic of period m, the sum of $\chi(n)$ over any complete system of m consecutive integers is equal to the sum of $\chi(x)$ over $x \in U_m$ (as $\chi(n) = 0$ when $\gcd(n,m) > 1$). Since $\chi \neq 1$, the First Orthogonality Relation (Proposition 1.3) gives

$$\sum_{n=u}^{u+m-1} \chi(n) = \sum_{x\in U_m} \chi(x) = 0 \quad \text{for every } u \geq 1.$$

Now, for any $1 \leq u \leq v$, write $v - u + 1 = qm + r$ with $q \geq 0$ and $0 \leq r < m$ by the division algorithm. Grouping the first qm terms of $A_{u,v}$ into q blocks of length m (each summing to 0), only at most $r < m$ terms remain, of which at most $\varphi(m)$ are coprime to m:

$$|A_{u,v}| = \left| \sum_{n=u+qm}^{v} \chi(n) \right| \leq \sum_{n=u+qm}^{v} |\chi(n)| \leq \varphi(m).$$

Thus $|A_{u,v}| \leq \varphi(m)$ for all $u \leq v$, so by Proposition 3.6 and Corollary 3.3, $L(s,\chi)$ converges and is holomorphic for $\Re(s) > 0$. □

Example (Leibniz's Formula for $L(1,\chi_4)$)**.** Take $m = 4$ and let χ_4 be the unique non-trivial character modulo 4. By Proposition 5.3, the series

$$L(s,\chi_4) = 1 - \frac{1}{3^s} + \frac{1}{5^s} - \frac{1}{7^s} + \cdots = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)^s}$$

converges for all $\Re(s) > 0$ and is holomorphic there. In particular, at $s = 1$, $L(1,\chi_4) = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \arctan(1) = \frac{\pi}{4} \neq 0$.

Remark. By Proposition 5.3, $L(1,\chi)$ is finite whenever $\chi \neq 1$. The *essential analytical heart* of Dirichlet's proof consists in showing that $L(1,\chi) \neq 0$ for every $\chi \neq 1$.

Towards this end, following Serre (Chapter VI, § 3.4), our next object of study is the product of all L-functions modulo m:

$$\zeta_m(s) := \prod_{\chi\in\hat{U}_m} L(s,\chi),$$

where the product is taken over all $\varphi(m)$ characters $\chi \in \hat{U}_m$. Let $p \nmid m$. By $\bar{p}$, denote the image of p in U_m. Let $\mathfrak{o}(p)$ denote the multiplicative order of $\bar{p} \in U_m$; that is, $\mathfrak{o}(p)$ is the smallest integer $f \geq 1$ such that $p^f \equiv 1 \pmod{m}$. Define $\mathfrak{g}(p) = \varphi(m)/\mathfrak{o}(p)$. It is the order of the quotient group $U_m/\langle\bar{p}\rangle$.

Lemma 5.4. For any prime $p \nmid m$ and all $z \in \mathbb{C}$,

$$\prod_{\chi\in\hat{U}_m} \big(1 - \chi(p)z\big) = \big(1 - z^{\mathfrak{o}(p)}\big)^{\mathfrak{g}(p)}.$$

Proof. Let $\mathfrak{o} = \mathfrak{o}(p)$ and $H = \langle\bar{p}\rangle \preceq U_m$, so $|H| = \mathfrak{o}$. The map $\rho_p : \hat{U}_m \to \mu_{\mathfrak{o}}$, $\chi \mapsto \chi(\bar{p})$, is the restriction $\hat{U}_m \to \hat{H}$ followed by the isomorphism $\hat{H} \cong \mu_{\mathfrak{o}}$ (H is cyclic), so it is onto by

Corollary 1.2. Its kernel has order $\varphi(m)/\mathfrak{o} = \mathfrak{g}(p)$, so each $w \in \mu_{\mathfrak{o}}$ equals $\chi(\bar{p})$ for exactly $\mathfrak{g}(p)$ characters. Therefore

$$\prod_{\chi \in \hat{U}_m} \big(1 - \chi(p)z\big) = \prod_{w \in \mu_{\mathfrak{o}}} (1 - wz)^{\mathfrak{g}(p)} = \big(1 - z^{\mathfrak{o}}\big)^{\mathfrak{g}(p)}. \qquad \square$$

Proposition 5.5. For $\Re(s) > 1$, one has the Euler product expansion

$$\zeta_m(s) = \prod_{\chi \in \hat{U}_m} L(s, \chi) = \prod_{p \nmid m} \frac{1}{\big(1 - p^{-\mathfrak{o}(p)s}\big)^{\mathfrak{g}(p)}}.$$

In particular, $\zeta_m(s)$ can be expanded as an ordinary Dirichlet series with non-negative integer coefficients, converging absolutely in the half-plane $\Re(s) > 1$.

Proof. For $\Re(s) > 1$, each $L(s, \chi)$ is given by the absolutely convergent Euler product $\prod_{p \nmid m}(1 - \chi(p)p^{-s})^{-1}$. Multiplying these finitely many ($\varphi(m)$) products together and applying Lemma 5.4 with $z = p^{-s}$ for each prime $p \nmid m$ gives

$$\zeta_m(s) = \prod_{\chi \in \hat{U}_m} \prod_{p \nmid m} \frac{1}{1 - \chi(p)p^{-s}} = \prod_{p \nmid m} \prod_{\chi \in \hat{U}_m} \frac{1}{1 - \chi(p)p^{-s}} = \prod_{p \nmid m} \frac{1}{\big(1 - p^{-\mathfrak{o}(p)s}\big)^{\mathfrak{g}(p)}}.$$

Expanding $(1 - p^{-\mathfrak{o}(p)s})^{-\mathfrak{g}(p)} = \sum_{k \geq 0} \binom{\mathfrak{g}(p)+k-1}{k} p^{-k\mathfrak{o}(p)s}$ and multiplying over $p \nmid m$ gives a Dirichlet series $\sum_n c_n n^{-s}$ with $c_n \in \mathbb{Z}_{\geq 0}$ and $c_1 = 1$. It converges absolutely for $\Re(s) > 1$, being a finite product of the absolutely convergent series $L(s, \chi)$. $\square$

Example (Explicit Computation of $\zeta_4(s)$)**.** Let $m = 4$, so $\varphi(4) = 2$ and $\zeta_4(s) = L(s, \chi_0)L(s, \chi_4)$. For an odd prime $p \nmid 4$:

- If $p \equiv 1 \pmod 4$, then $\bar{p} = \overline{1}$ in U_4, so $\mathfrak{o}(p) = 1$ and $\mathfrak{g}(p) = 2/1 = 2$. The Euler factor at p is $(1 - p^{-s})^{-2} = \sum_{k=0}^{\infty}(k + 1)p^{-ks}$.
- If $p \equiv 3 \pmod 4$, then $\bar{p} = \overline{3}$ has order $\mathfrak{o}(p) = 2$ in U_4, so $\mathfrak{g}(p) = 2/2 = 1$. The Euler factor at p is $(1 - p^{-2s})^{-1} = \sum_{k=0}^{\infty} p^{-2ks}$.

Hence

$$\zeta_4(s) = \prod_{p \equiv 1 \pmod 4} \frac{1}{(1 - p^{-s})^2} \prod_{p \equiv 3 \pmod 4} \frac{1}{1 - p^{-2s}} = \sum_{\substack{n=1 \\ n \text{ odd}}}^{\infty} \frac{r_2(n)/4}{n^s},$$

where $\frac{1}{4}r_2(n) = \sum_{d|n} \chi_4(d)$ and $r_2(n) = \#\{(a, b) \in \mathbb{Z}^2 : a^2 + b^2 = n\}$. Up to its Euler factor at 2, ζ_4 is the Dedekind zeta function of $\mathbb{Q}(i)$.

We are now ready to prove the central theorem of Chapter VI, §3.4 of Serre (Theorem 1), combining the analytic continuation of $L(s, \chi)$, the positivity of the Dirichlet coefficients of $\zeta_m(s)$, and Landau's Theorem (Theorem 3.4).

Theorem 5.6 (Non-vanishing of $L(1, \chi)$)**.** (a) $\zeta_m(s)$ has a simple pole at $s = 1$.

(b) $L(1, \chi) \neq 0$ for all $\chi \neq 1$.

Proof. By Proposition 5.2, $L(s, 1)$ is holomorphic on $\Re(s) > 0$ except for a simple pole at $s = 1$, and by Proposition 5.3 every $L(s, \chi)$ with $\chi \neq 1$ is holomorphic on $\Re(s) > 0$. So (b) implies (a): if no $L(1, \chi)$ vanishes, then $\zeta_m = L(s, 1) \prod_{\chi \neq 1} L(s, \chi)$ has a simple pole at $s = 1$.

It remains to prove (b). Suppose $L(1,\chi_0)=0$ for some $\chi_0 \neq 1$. Then this zero cancels the pole of $L(s,1)$, so $\zeta_m(s)$ is holomorphic on the whole half-plane $\Re(s)>0$. By Proposition 5.5, $\zeta_m(s) = \sum_n c_n n^{-s}$ with $c_n \geq 0$, convergent for $\Re(s)>1$. Let $\rho \leq 1$ be its abscissa of convergence. If $\rho > 0$, then ζ_m is holomorphic near the real point ρ, and Landau's theorem (Theorem 3.4) gives convergence on $\Re(s) > \rho - \varepsilon$, contradicting ρ. Hence the series converges for all $\Re(s) > 0$.

This is impossible. Let $p \nmid m$, let $s>0$ be real, and put $u = p^{-\mathfrak{o}(p)s}$. Since $\mathfrak{o}(p)\mathfrak{g}(p) = \varphi(m)$,

$$(1-u)^{-\mathfrak{g}(p)} = (1+u+u^2+\cdots)^{\mathfrak{g}(p)} \;\geq\; 1 + u^{\mathfrak{g}(p)} + u^{2\mathfrak{g}(p)} + \cdots = \sum_{k\geq 0} p^{-k\varphi(m)s}$$

coefficientwise (all coefficients are non-negative). Multiplying over $p \nmid m$ gives $c_{n^{\varphi(m)}} \geq 1$ whenever $\gcd(n,m)=1$. At $s = 1/\varphi(m)$ this yields

$$\sum_N \frac{c_N}{N^s} \;\geq \sum_{\gcd(n,m)=1} \frac{1}{n} = \infty,$$

because the right-hand sum contains $\sum_k \frac{1}{km+1}$. So the series diverges at $s = 1/\varphi(m) > 0$, a contradiction. Hence $L(1,\chi) \neq 0$ for all $\chi \neq 1$. □

6 Density and Dirichlet's Theorem

6.1 Analytic Density and the Statement of Dirichlet's Theorem

Let P be the set of all prime numbers. Recall from Corollary 4.5 that as $s \to 1^+$ along the real axis ($s>1$),

$$\sum_{p\in P} \frac{1}{p^s} \sim \log\left(\frac{1}{s-1}\right).$$

This motivates measuring the "size" or proportion of a subset $A \subseteq P$ by comparing the blow-up rate of $\sum_{p\in A} p^{-s}$ as $s \to 1^+$ to that of $\sum_{p\in P} p^{-s} \sim \log \frac{1}{s-1}$ (Serre, Chapter VI, §4.1).

Definition 6.1 (Analytic / Dirichlet Density)**.** Let $A \subseteq P$ be a set of prime numbers. We say that A has *analytic density* (or *Dirichlet density*) equal to $k \in \mathbb{R}$ if

$$\lim_{s\to 1^+} \frac{\displaystyle\sum_{p\in A} \frac{1}{p^s}}{\log\left(\dfrac{1}{s-1}\right)} = k.$$

Clearly, if the analytic density k of A exists, then $0 \leq k \leq 1$. Moreover:

- If A is *finite*, then $\sum_{p\in A} p^{-s} \to \sum_{p\in A} p^{-1} < \infty$ as $s \to 1^+$, while $\log \frac{1}{s-1} \to +\infty$, so $\mathrm{dens}(A) = 0$. Consequently, **any subset $A \subseteq P$ with strictly positive analytic density $k>0$ is necessarily infinite.**
- Analytic density is finitely additive: if $A_1, A_2 \subseteq P$ are disjoint and have densities k_1, k_2, then $A_1 \cup A_2$ has density $k_1 + k_2$.

Dirichlet's theorem on primes in arithmetic progressions can now be stated in its sharp quantitative form (Serre, VI.4.1, Theorem 2):

Theorem 6.2 (Dirichlet's Theorem on Arithmetic Progressions)**.** Let $m \geq 1$ and let $a \in \mathbb{Z}$ be such that $\gcd(a, m) = 1$. Let

$$P_a = \{p \in P : p \equiv a \pmod{m}\}$$

be the set of prime numbers congruent to a modulo m. Then P_a has analytic density $\frac{1}{\varphi(m)}$:

$$\lim_{s \to 1^+} \frac{\sum_{p \equiv a \pmod{m}} \frac{1}{p^s}}{\log\left(\frac{1}{s-1}\right)} = \frac{1}{\varphi(m)}.$$

In other words, the prime numbers are *equidistributed* among the $\varphi(m)$ invertible residue classes modulo m.

Corollary 6.3. For any coprime integers a and $m \geq 1$, there exist infinitely many prime numbers p such that $p \equiv a \pmod{m}$.

Proof. Since $\frac{1}{\varphi(m)} > 0$ and any finite set of primes has density 0, P_a must be infinite. □

6.2 Logarithms of L-functions and the Proof of Dirichlet's Theorem

To prove Theorem 6.2, following Serre (Chapter VI, §4.2–4.3), we associate to each character $\chi \in \hat{U}_m$ the prime Dirichlet series

$$f_\chi(s) := \sum_{p \nmid m} \frac{\chi(p)}{p^s},$$

which converges absolutely for $\Re(s) > 1$.

Lemma 6.4. If $\chi = 1$, then $f_1(s) \sim \log\left(\frac{1}{s-1}\right)$ as $s \to 1^+$.

Proof. The series $f_1(s) = \sum_{p \nmid m} p^{-s}$ differs from $\sum_{p \in P} p^{-s}$ only by the finite sum $\sum_{p \mid m} p^{-s}$, which remains bounded (in fact approaches $\sum_{p \mid m} p^{-1} < \infty$) as $s \to 1^+$. The result therefore follows from Corollary 4.5. □

Lemma 6.5. If $\chi \neq 1$, then $f_\chi(s)$ remains bounded as $s \to 1^+$.

Proof. For $\Re(s) > 1$ put

$$\log L(s, \chi) := \sum_{p \nmid m} \sum_{k \geq 1} \frac{\chi(p)^k}{k\, p^{ks}}.$$

This converges absolutely (as $|\chi(p)p^{-s}| < 1$) and, by the Euler product and $-\log(1-\alpha) = \sum_{k \geq 1} \alpha^k/k$, satisfies $\exp\big(\log L(s,\chi)\big) = L(s,\chi)$. Write $\log L(s,\chi) = f_\chi(s) + F_\chi(s)$ with $F_\chi(s) = \sum_{p \nmid m} \sum_{k \geq 2} \frac{\chi(p)^k}{k\, p^{ks}}$. For real $s > 1$, $|F_\chi(s)| \leq \Theta(s) \leq 1$ by the estimate in the proof of Corollary 4.5.

By Proposition 5.3 and Theorem 5.6(b), $L(s,\chi) \to L(1,\chi) \neq 0$ as $s \to 1^+$. Choose $r > 0$ such that $L(s,\chi)$ stays in a disc not containing 0 for $s \in (1, 1+r)$. On that disc a continuous

logarithm exists and is bounded, and two continuous logarithms of $L(s,\chi)$ on $(1,1+r)$ differ by a constant in $2\pi i\mathbb{Z}$. Hence $\log L(s,\chi)$ is bounded on $(1,1+r)$, and so is $f_\chi = \log L(\cdot,\chi) - F_\chi$. $\square$

Finally, we use the Second Orthogonality Relation (Corollary 1.3) to extract the sum over primes $p \in P_a$ from the family $\{f_\chi(s)\}_{\chi\in\hat{U}_m}$. For $\gcd(a,m)=1$, define

$$g_a(s) := \sum_{p\in P_a} \frac{1}{p^s} = \sum_{p\equiv a \pmod m} \frac{1}{p^s} \quad (\Re(s)>1).$$

Lemma 6.6 (Fourier Inversion for Primes in P_a)**.** For $\Re(s)>1$, one has

$$g_a(s) = \frac{1}{\varphi(m)} \sum_{\chi\in\hat{U}_m} \chi(a)^{-1} f_\chi(s).$$

Proof. Substituting the definition $f_\chi(s) = \sum_{p\nmid m} \chi(p)p^{-s}$ and interchanging the finite sum over $\chi \in \hat{U}_m$ with the absolutely convergent sum over $p \nmid m$, we obtain

$$\sum_{\chi\in\hat{U}_m} \chi(a)^{-1} f_\chi(s) = \sum_{p\nmid m} \left(\sum_{\chi\in\hat{U}_m} \chi(a)^{-1}\chi(p) \right) \frac{1}{p^s} = \sum_{p\nmid m} \left(\sum_{\chi\in\hat{U}_m} \chi(a^{-1}p) \right) \frac{1}{p^s}.$$

By Corollary 1.3 applied to the element $x = \overline{a}^{-1}\overline{p} \in U_m$ (where $|U_m| = \varphi(m)$), we have

$$\sum_{\chi\in\hat{U}_m} \chi(a^{-1}p) = \begin{cases} \varphi(m) & \text{if } a^{-1}p \equiv 1 \pmod m \Leftrightarrow p \equiv a \pmod m, \\ 0 & \text{otherwise.} \end{cases}$$

Hence the right-hand side simplifies to $\varphi(m)\sum_{p\equiv a \pmod m} p^{-s} = \varphi(m) g_a(s)$. Dividing by $\varphi(m)$ proves the lemma. $\square$

Proof of Theorem 6.2 (Dirichlet's Theorem). By Lemma 6.6, separating the principal character $\chi = 1$ (for which $1(a)^{-1} = 1$) from the $\varphi(m)-1$ non-trivial characters $\chi \neq 1$, we can write for $s>1$:

$$g_a(s) = \frac{1}{\varphi(m)} f_1(s) + \frac{1}{\varphi(m)} \sum_{\substack{\chi\in\hat{U}_m \\ \chi\neq 1}} \chi(a)^{-1} f_\chi(s).$$

Dividing both sides by $\log\left(\frac{1}{s-1}\right)$ and taking the limit as $s \to 1^+$:

- By Lemma 6.4, $\lim_{s\to 1^+} \frac{f_1(s)}{\log(1/(s-1))} = 1$, so the $\chi = 1$ term contributes $\frac{1}{\varphi(m)}$;
- By Lemma 6.5, for each $\chi \neq 1$, $f_\chi(s) = O(1)$ remains bounded as $s \to 1^+$, while $\log\left(\frac{1}{s-1}\right) \to +\infty$, so $\lim_{s\to 1^+} \frac{f_\chi(s)}{\log(1/(s-1))} = 0$.

Therefore,

$$\lim_{s\to 1^+} \frac{g_a(s)}{\log\left(\frac{1}{s-1}\right)} = \frac{1}{\varphi(m)} \cdot 1 + 0 = \frac{1}{\varphi(m)},$$

completing the proof of Dirichlet's theorem. $\square$

Example (Primes Congruent to 1 and 3 modulo 4)**.** Let $m = 4$, where $U_4 = \{\overline{1}, \overline{3}\}$ and $\hat{U}_4 = \{1, \chi_4\}$. Here

$$f_1(s) = \sum_{p \text{ odd}} \frac{1}{p^s} = \sum_{p \equiv 1 \pmod 4} \frac{1}{p^s} + \sum_{p \equiv 3 \pmod 4} \frac{1}{p^s}, \qquad f_{\chi_4}(s) = \sum_{p \equiv 1 \pmod 4} \frac{1}{p^s} - \sum_{p \equiv 3 \pmod 4} \frac{1}{p^s}.$$

Hence Lemma 6.6 takes the transparent form

$$g_1(s) = \frac{f_1(s) + f_{\chi_4}(s)}{2}, \qquad g_3(s) = \frac{f_1(s) - f_{\chi_4}(s)}{2}.$$

Because $L(1, \chi_4) = \frac{\pi}{4} \neq 0$, the difference $f_{\chi_4}(s) = g_1(s) - g_3(s)$ remains bounded as $s \to 1^+$, while the sum $f_1(s) = g_1(s) + g_3(s) \sim \log \frac{1}{s-1}$ diverges to $+\infty$. Consequently, both P_1 and P_3 have analytic density $\frac{1}{2}$.

6.3 An Application to Quadratic Residues

Combining Theorem 6.2 with Proposition 1.5 yields a striking equidistribution theorem for quadratic residues (Serre, Chapter VI, §4.4, Proposition 14):

Proposition 6.7. Let $a \in \mathbb{Z}$ be an integer which is not a perfect square. Then the set of prime numbers p such that $\left(\frac{a}{p}\right) = 1$ (i.e., a is a quadratic residue modulo p) has analytic density $\frac{1}{2}$.

Proof. Since $a \neq 0$ (as $0 = 0^2$ is a square), we may write $a = a_0 c^2$ where a_0 is square-free and $c \geq 1$. For every prime $p \nmid 2ac$, we have $\left(\frac{a}{p}\right) = \left(\frac{a_0}{p}\right)$, and removing the finitely many primes dividing $2ac$ does not change the analytic density. Thus we may assume without loss of generality that a is square-free and $a \neq 1$.

Let $m = 4|a|$, and let $\chi_a : U_m \to \{\pm 1\}$ be the quadratic character modulo m constructed in Proposition 1.5. Since $a \neq 1$, χ_a is a non-trivial surjective homomorphism $U_m \to \{\pm 1\}$. Thus its kernel $H = \ker(\chi_a) = \{\bar{x} \in U_m : \chi_a(x) = 1\}$ is a subgroup of index $[U_m : H] = 2$ in U_m, so $|H| = \frac{1}{2}\varphi(m)$.

For any prime $p \nmid m$, Proposition 1.5 gives $\left(\frac{a}{p}\right) = \chi_a(p)$, so $\left(\frac{a}{p}\right) = 1 \Leftrightarrow \bar{p} \in H$. Therefore, up to the finitely many primes dividing m, the set $\left\{p \in P : \left(\frac{a}{p}\right) = 1\right\}$ is the disjoint union of the $|H| = \frac{1}{2}\varphi(m)$ sets P_b for $\bar{b} \in H$. By Theorem 6.2 and finite additivity of analytic density, its density is

$$\sum_{\bar{b} \in H} \operatorname{dens}(P_b) = |H| \cdot \frac{1}{\varphi(m)} = \frac{\varphi(m)}{2} \cdot \frac{1}{\varphi(m)} = \frac{1}{2}. \qquad \square$$

Corollary 6.8 (Local-Global Principle for Squares)**.** Let $a \in \mathbb{Z}$. If the congruence $X^2 - a \equiv 0 \pmod p$ has a solution for almost all $p \in P$ (i.e., for all but finitely many primes p), then a is a square in $\mathbb{Z}$.

Proof. If a were not a square in $\mathbb{Z}$, then by Proposition 6.7 the set of primes p where $X^2 - a \equiv 0 \pmod p$ is solvable would have density $\frac{1}{2}$, whereas the hypothesis that it is solvable for all but finitely many primes implies its density is $1 \neq \frac{1}{2}$, a contradiction. $\square$

Remark (Frobenius and Chebotarev Density Theorems)**.** Serre notes (VI.4.4) that this extends to any irreducible $f \in \mathbb{Z}[X]$: if K is its splitting field and $N = [K : \mathbb{Q}]$, the set of primes p for which f splits completely mod p has analytic density $1/N$.

6.4 Natural Density vs. Analytic Density

Following Serre (Chapter VI, §4.5), there is a more intuitive notion of density for a subset $A \subseteq P$:

Definition 6.9 (Natural Density)**.** A subset $A \subseteq P$ has *natural density* $k \in \mathbb{R}$ if

$$\lim_{n\to\infty} \frac{\mathrm{Card}\{p \in A : p \le n\}}{\mathrm{Card}\{p \in P : p \le n\}} = k.$$

Using Abel's summation by parts (Lemma 2.3) and the Prime Number Theorem ($\pi(n) = \mathrm{Card}\{p \in P : p \le n\} \sim \frac{n}{\log n}$), one can prove that if A has natural density k, then its analytic density exists and is also equal to k.

The converse, however, is false: there exist subsets $A \subseteq P$ which possess an analytic density but fail to have a natural density. Serre gives the striking example (communicated by E. Bombieri) of the set $P^1 \subseteq P$ of prime numbers whose first digit in base 10 is 1:

$$P^1 = \bigcup_{k=0}^{\infty} \{p \in P : 10^k \le p < 2 \cdot 10^k\}.$$

By the Prime Number Theorem, $\mathrm{Card}\{p \in P^1 : p \le n\}/\pi(n)$ tends to $\frac{5}{9}$ along $n = 2 \cdot 10^k - 1$ and to $\frac{1}{9}$ along $n = 10^{k+1} - 1$, so P^1 has no natural density. But every decade $[10^k, 10^{k+1})$ has the same logarithmic weight, and P^1 has analytic density $\log_{10} 2 \approx 0.301$ (Benford's law).

For arithmetic progressions this pathology does not occur: the non-vanishing of $L(s, \chi)$ on the line $\Re(s) = 1$, together with a Tauberian theorem, gives the prime number theorem for arithmetic progressions: for $\gcd(a, m) = 1$,

$$\mathrm{Card}\{p \in P_a : p \le x\} \sim \frac{1}{\varphi(m)} \frac{x}{\log x} \quad \text{as } x \to \infty.$$

7 Modular Forms

Following Chapter VII (§§1–3) of Serre, we study holomorphic functions on $\mathbb{H} = \{\Im z > 0\}$ that are symmetric under $\mathrm{SL}_2(\mathbb{Z})$.

7.1 The modular group

$\mathrm{SL}_2(\mathbb{R})$ acts on $\mathbb{H}$ by $gz = \frac{az+b}{cz+d}$, and

$$\Im(gz) = \frac{\Im(z)}{|cz+d|^2}, \qquad \frac{d(gz)}{dz} = (cz+d)^{-2}. \tag{2}$$

The kernel of the action is $\{\pm 1\}$. Put $G = \mathrm{PSL}_2(\mathbb{Z}) = \mathrm{SL}_2(\mathbb{Z})/\{\pm 1\}$, the *modular group*, and

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad Sz = -\frac{1}{z}, \quad Tz = z + 1,$$

with $S^2 = (ST)^3 = 1$ in G. Let $D = \{z \in \mathbb{H} : |z| \ge 1,\ |\Re z| \le \frac{1}{2}\}$ and $\rho = e^{2\pi i/3}$, so $-\bar{\rho} = e^{\pi i/3}$ (Figure 5).

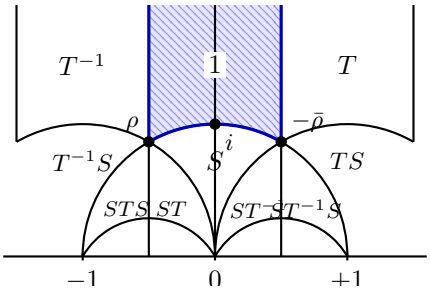


Figure 5: The fundamental domain D (hatched region labeled 1) of the modular group $G = \mathrm{PSL}_2(\mathbb{Z})$ and its tessellation transforms in the upper half-plane $\mathbb{H}$ (cf. Serre, Chapter VII, Fig. 1).

Theorem 7.1 (Serre, VII.1.2)**.** (1) Every $z \in \mathbb{H}$ is G-equivalent to a point of D.

(2) Two distinct points of D are G-equivalent iff $\Re z = \pm\frac{1}{2}$ and $z' = z \mp 1$, or $|z| = 1$ and $z' = -1/z$.

(3) The stabilizer $I(z)$ of $z \in D$ is trivial except $I(i) = \langle S \rangle$ (order 2), $I(\rho) = \langle ST \rangle$ and $I(-\bar{\rho}) = \langle TS \rangle$ (order 3).

Moreover G is generated by S and T.

Proof. Let $G' = \langle S, T \rangle$. (1) For $z \in \mathbb{H}$ there are only finitely many $(c, d) \in \mathbb{Z}^2$ with $|cz + d| \leq 1$, so by (2) $\Im(gz)$ attains a maximum on $G'z$, at gz say. Translating, we may assume $|\Re gz| \leq \frac{1}{2}$. If $|gz| < 1$ then $\Im(Sgz) > \Im(gz)$, a contradiction; so $gz \in D$.

(2),(3) Let $z \in D$ and $gz \in D$; replacing (z, g) by (gz, g^{-1}) we may assume $|cz + d| \leq 1$. As $\Im z \geq \frac{\sqrt{3}}{2}$, $|c| \leq 1$; we may take $c \in \{0, 1\}$. If $c = 0$, g is a translation, so $g = 1$ or $gz = z \pm 1$ with $\Re z = \mp\frac{1}{2}$. If $c = 1$, then $|z + d| \leq 1$ forces $|d| \leq 1$. For $d = 0$: $|z| = 1$ and $gz = a - \bar{z}$, so $gz = -1/z$ ($a = 0$), or $gz = z$ with $g = TS$, $T^{-1}S$ (at $-\bar{\rho}$, ρ). For $d = \pm 1$: $z = \rho$ (resp. $-\bar{\rho}$) and $g \in \{ST,\ TST\}$ (resp. $\{ST^{-1},\ T^{-1}ST^{-1}\}$), where ST fixes ρ and ST^{-1} fixes $-\bar{\rho}$. This proves (2) and (3).

Generators: for $g \in G$ and $z_0 = 2i$, (1) gives $g' \in G'$ with $g'gz_0 \in D$; by (2),(3), $g'gz_0 = z_0$ and $g'g = 1$, so $g \in G'$. □

7.2 Modular functions

Definition 7.2. A meromorphic f on $\mathbb{H}$ is *weakly modular of weight* $2k$ if $f(z) = (cz + d)^{-2k} f(gz)$ for all $g \in \mathrm{SL}_2(\mathbb{Z})$, i.e. $f(z)(dz)^k$ is G-invariant (odd weights are impossible, as -1 acts trivially). By Theorem 7.1 this holds iff $f(z + 1) = f(z)$ and $f(-1/z) = z^{2k} f(z)$. Then $f(z) = \tilde{f}(q)$, $q = e^{2\pi i z}$, with $\tilde{f}$ meromorphic on $0 < |q| < 1$. If $\tilde{f}$ is meromorphic (resp. holomorphic) at $q = 0$, f is *meromorphic* (resp. *holomorphic*) *at* ∞, with $f(\infty) = \tilde{f}(0)$. A *modular function* is weakly modular and meromorphic at ∞; a *modular form* is moreover holomorphic on $\mathbb{H}$ and at ∞; a *cusp form* is a modular form with $f(\infty) = 0$.

Lattices give a conceptual picture (Serre, VII.2.2). Let M be the set of $(\omega_1, \omega_2) \in (\mathbb{C}^\times)^2$ with $\Im(\omega_1/\omega_2) > 0$. Then $M/\mathrm{SL}_2(\mathbb{Z})$ is the set $\mathcal{R}$ of lattices of $\mathbb{C}$ and $\mathcal{R}/\mathbb{C}^\times \cong \mathbb{H}/G$ via $(\omega_1, \omega_2) \mapsto \omega_1/\omega_2$. A lattice function $F(\lambda\Gamma) = \lambda^{-2k} F(\Gamma)$ corresponds to the weakly modular function $f(z) = F(\Gamma(z, 1))$ of weight $2k$.

Lemma 7.3. For a lattice $\Gamma \subset \mathbb{C}$, $\sum'_{\gamma\in\Gamma} |\gamma|^{-\sigma} < \infty$ for $\sigma > 2$.

Proof. For a basis ω_1, ω_2, there are $8N$ lattice points with $\max(|m|, |n|) = N$, each with $|\gamma| \geq CN$; so the sum is $\leq \sum_N 8N(CN)^{-\sigma} < \infty$. □

For $k \geq 2$ the *Eisenstein series* is $G_k(\Gamma) = \sum'_{\gamma\in\Gamma} \gamma^{-2k}$, with

$$G_k(z) = \sum_{(m,n)\neq(0,0)} \frac{1}{(mz+n)^{2k}}. \tag{3}$$

Proposition 7.4. G_k is a modular form of weight $2k$, and $G_k(\infty) = 2\zeta(2k)$.

Proof. It is weakly modular of weight $2k$, being a lattice function. On a compact $K \subseteq \mathbb{H}$, $|mz+n|^2 \geq c_K(m^2+n^2)$ for some $c_K > 0$, so by Lemma 7.3 the series converges normally: G_k is holomorphic. On D (enough, by periodicity) $|mz+n|^2 \geq m^2 - |mn| + n^2 \geq \frac{1}{2}(m^2+n^2)$, so the series is uniformly dominated and we may let $\Im z \to \infty$ termwise; only the terms $m = 0$ survive, giving $\sum_{n\neq 0} n^{-2k} = 2\zeta(2k)$. This also shows G_k is bounded at ∞, hence holomorphic there. □

Example. By $\zeta(4) = \frac{\pi^4}{90}$ and $\zeta(6) = \frac{\pi^6}{945}$, the forms $g_2 = 60G_2$ and $g_3 = 140G_3$ have $g_2(\infty) = \frac{4}{3}\pi^4$, $g_3(\infty) = \frac{8}{27}\pi^6$. Hence

$$\Delta = g_2^3 - 27g_3^2 \tag{4}$$

is a cusp form of weight 12. It is the discriminant of the cubic $4x^3 - g_2x - g_3$ (up to a factor 16), where $(x, y) = (\wp_\Gamma, \wp'_\Gamma)$ embeds $\mathbb{C}/\Gamma$ in $\mathbb{P}^2$.

7.3 The space of modular forms

For a non-zero modular function f and $p \in \mathbb{H}$ let $v_p(f)$ be the order of f at p (it depends only on the G-orbit of p), and $v_\infty(f)$ the order of $\tilde{f}$ at $q = 0$.

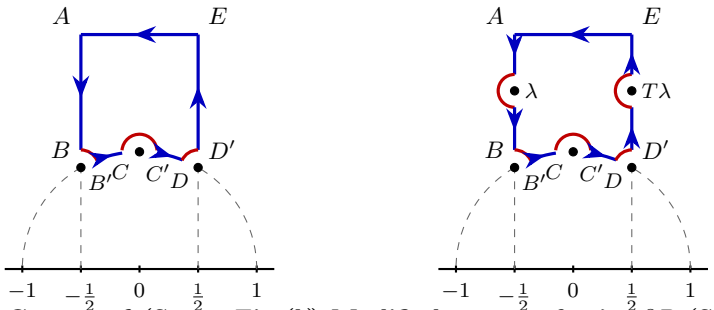
Theorem 7.5 (Valence formula)**.** For a non-zero modular function f of weight $2k$,

$$v_\infty(f) + \tfrac{1}{2}v_i(f) + \tfrac{1}{3}v_\rho(f) + \sum_{p\in\mathbb{H}/G}^{*} v_p(f) = \frac{k}{6}, \tag{5}$$

where $\sum^*$ runs over the classes other than those of i and ρ.

Proof. f has finitely many zeros and poles in $\mathbb{H}/G$, since it has none for $\Im z$ large and the rest of D is compact. Assume first there are none on ∂D except possibly at $i, \rho, -\bar{\rho}$, and integrate $\frac{1}{2\pi i}\frac{df}{f}$ over the contour $\mathcal{C}$ of Figure 6(a); by the residue theorem the integral is $\sum^* v_p(f)$ (as $\varepsilon \to 0$). The pieces contribute:

- the top segment EA: $-v_\infty(f)$ (in $q = e^{2\pi iz}$ it is a circle traversed negatively);
- the small arcs at ρ, $-\bar{\rho}$ and i (angles $\frac{\pi}{3}$, $\frac{\pi}{3}$, π): $-\frac{1}{6}v_\rho(f)$, $-\frac{1}{6}v_\rho(f)$ and $-\frac{1}{2}v_i(f)$;
- the vertical sides: they cancel, since $f(z+1) = f(z)$;



(a) Contour $\mathcal{C}$ (Serre, Fig. 2) **(b) Modified contour for $\lambda \in \partial D$ (Serre, Fig. 3)**

Figure 6: The contour of integration $\mathcal{C}$ along the truncated fundamental domain used to prove the Valence Formula (Theorem 7.5): (a) when f has no zeros or poles on $\partial D \setminus \{i, \rho, -\bar{\rho}\}$; (b) modified by T-translates of small semi-circular indentations when f has a zero or pole λ on the vertical boundary.

- the arcs $B'C$ and $C'D$: S maps one onto the other reversed, and $d\log f(Sz) = 2k\frac{dz}{z} + d\log f(z)$, so together they give $\frac{1}{2\pi i}\int_{B'}^{C}(-2k)\frac{dz}{z} \to -2k\cdot(-\frac{1}{12}) = \frac{k}{6}$.

Summing gives (5). If f has a zero or pole λ on ∂D, indent the contour around λ and around its image under T (resp. S) by translates (resp. S-images) of each other, so the integrals along matching sides still cancel (Figure 6(b)). □

Let M_k (resp. M_k^0) be the space of modular forms (resp. cusp forms) of weight $2k$. Then M_k^0 is the kernel of $f \mapsto f(\infty)$, so $\dim M_k \le 1 + \dim M_k^0$, and

$$M_k = M_k^0 \oplus \mathbb{C}G_k \qquad (k \ge 2). \tag{6}$$

Theorem 7.6. (i) $M_k = 0$ for $k < 0$ and $k = 1$.

(ii) For $k = 0, 2, 3, 4, 5$, M_k has basis $1, G_2, G_3, G_4, G_5$ respectively, and $M_k^0 = 0$.

(iii) Multiplication by Δ is an isomorphism $M_{k-6} \xrightarrow{\sim} M_k^0$ for all k.

Proof. For $0 \ne f \in M_k$ all orders in (5) are ≥ 0, so $k \ge 0$, and $k \ne 1$ since $\frac{1}{6}$ is not of the form $n + \frac{n'}{2} + \frac{n''}{3}$ with $n, n', n'' \in \mathbb{Z}_{\ge 0}$. This is (i).

For G_2 ($k = 2$) and G_3 ($k = 3$) the right side is $\frac{1}{3}$, resp. $\frac{1}{2}$, forcing $v_\rho(G_2) = 1$ with no other zero, and $v_i(G_3) = 1$ with no other zero. Hence $G_2(i) \ne 0$ and $\Delta(i) = g_2(i)^3 \ne 0$, so $\Delta \ne 0$. As $\Delta \in M_6^0$ and the right side is 1 for $k = 6$, we get $v_\infty(\Delta) = 1$ and $v_p(\Delta) = 0$ for $p \in \mathbb{H}$. So for $f \in M_k^0$, f/Δ is holomorphic on $\mathbb{H}$ and at ∞, of weight $2(k-6)$: it lies in M_{k-6}. This is (iii).

For $k \le 5$, (i) and (iii) give $M_k^0 \cong M_{k-6} = 0$, so $\dim M_k \le 1$; since $G_k(\infty) \ne 0$, $1, G_2, \dots, G_5$ are bases. This is (ii). □

Corollary 7.7. $\dim M_k = \lfloor k/6 \rfloor$ if $k \equiv 1 \pmod 6$ and $\lfloor k/6 \rfloor + 1$ otherwise. A basis of M_k is $\{G_2^\alpha G_3^\beta : 2\alpha + 3\beta = k\}$; so $\bigoplus_k M_k = \mathbb{C}[G_2, G_3]$.

Proof. By (6) and (iii), $\dim M_k = 1 + \dim M_{k-6}$ for $k \ge 6$, which with (ii) gives the dimension formula. Spanning, by induction on k: if $g = G_2^\gamma G_3^\delta$ has weight $2k$, then $g(\infty) \ne 0$, so $f - \lambda g \in M_k^0 = \Delta M_{k-6}$ for suitable λ, and $\Delta = (60G_2)^3 - 27(140G_3)^2$. Independence: a relation among

the monomials of weight $2k$ gives a polynomial relation for G_2^3/G_3^2, which is non-constant (it vanishes at ρ and has a pole at i), so it is trivial. □

Example. Since $\dim M_4 = \dim M_5 = 1$, $G_4 = \frac{2\zeta(8)}{(2\zeta(4))^2} G_2^2 = \frac{3}{7} G_2^2$ and $G_5 = \frac{5}{11} G_2 G_3$. With $E_k = G_k/2\zeta(2k)$ and the q-expansions $E_2 = 1 + 240 \sum \sigma_3(n) q^n$, $E_4 = 1 + 480 \sum \sigma_7(n) q^n$ (Serre, VII.4.2), the identity $E_4 = E_2^2$ yields

$$\sigma_7(n) = \sigma_3(n) + 120 \sum_{m=1}^{n-1} \sigma_3(m)\sigma_3(n-m).$$

The modular invariant

Since g_2^3 and Δ both lie in M_6, $j = 1728\, g_2^3/\Delta$ is a modular function of weight 0.

Proposition 7.8. j is holomorphic on $\mathbb{H}$, has a simple pole at ∞ ($j = q^{-1} + 744 + 196884q + \cdots$), and induces a bijection $\mathbb{H}/G \to \mathbb{C}$.

Proof. Since Δ has no zeros on $\mathbb{H}$ and $v_\infty(\Delta) = 1$, $v_\infty(g_2^3) = 0$, the first claims follow. For $\lambda \in \mathbb{C}$, $f_\lambda = 1728 g_2^3 - \lambda\Delta \in M_6$ has $f_\lambda(\infty) \neq 0$ and $j(z) = \lambda \Leftrightarrow f_\lambda(z) = 0$. By (5), $\frac{1}{2} v_i + \frac{1}{3} v_\rho + \sum^* v_p = 1$, whose non-negative integer solutions are: one simple zero in a class other than i, ρ; or $v_i = 2$ (when $\lambda = j(i) = 1728$); or $v_\rho = 3$ (when $\lambda = j(\rho) = 0$). In each case there is exactly one zero in $\mathbb{H}/G$. □

Proposition 7.9. For f meromorphic on $\mathbb{H}$, the following are equivalent: (i) f is a modular function of weight 0; (ii) f is a quotient of modular forms of equal weight; (iii) $f \in \mathbb{C}(j)$.

Proof. (iii)⇒(ii)⇒(i) is clear. For (i)⇒(iii), multiplying f by $\prod (j - j(p_i))^{m_i}$ over its poles p_i (of orders m_i) removes all poles in $\mathbb{H}$, so we may assume f holomorphic on $\mathbb{H}$. For large n, $\Delta^n f \in M_{6n}$, a combination of monomials $G_2^\alpha G_3^\beta$ with $2\alpha + 3\beta = 6n$. Then $\alpha = 3a$, $\beta = 2b$, and $G_2^{3a} G_3^{2b}/\Delta^{a+b}$ is a polynomial in j, since $g_2^3/\Delta = \frac{j}{1728}$ and $27 g_3^2/\Delta = \frac{j}{1728} - 1$. So $f \in \mathbb{C}[j]$ (after undoing the first step, $f \in \mathbb{C}(j)$). □

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